

Discovery Length-7 Prime Chains under a 4-adic Collatz-like Map

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A research note on prime-generating behavior in a 4-adic Collatz-like map:

$$f(x) = (5x \pm 1)/4, \text{ with the sign determined by } x \bmod 4$$

Discovered Length-7 Prime Chains

Chain 1:

455,847,361,774,092,289
569,809,202,217,615,361
712,261,502,772,019,201
890,326,878,465,024,001
1,112,908,598,081,280,001
1,391,135,747,601,600,001
1,738,919,684,502,000,001

Chain 2:

559,188,056,938,807,297
698,985,071,173,509,121
873,731,338,966,886,401
1,092,164,173,708,608,001
1,365,205,217,135,760,001
1,706,506,521,419,700,001
2,133,133,151,774,625,001